

Aether — Universal Runtime

Portability

Business Value Proposition

Deploy once. Move workloads across containers, Kubernetes, VMs, and bare metal without rewriting infrastructure.

Written in Rust • Open Source • Production Ready

The Problem

Modern infrastructure teams juggle multiple runtimes, each with its own toolchain, configuration language, and operational model. The result is fragmentation, risk, and waste.



Vendor Lock-In

Teams commit to a single runtime early and pay the price later. Moving workloads between Kubernetes and bare-metal requires rewriting deployment manifests, CI/CD pipelines, and monitoring integrations from scratch.



Runtime Sprawl

Containers on dev laptops, Kubernetes in staging, VMs in production, bare-metal for GPU workloads. Each runtime demands its own experts, tooling, and runbooks, multiplying operational complexity.



Migration Complexity

Migrating a workload from Podman to Kubernetes means rewriting YAML, reconfiguring networking, updating secrets management, and revalidating health checks. A single migration can take weeks of engineering effort.



Operational Overhead

Four runtimes means four sets of dashboards, four alerting configurations, four log formats, and four on-call playbooks. Teams spend more time on tooling than on delivering business value.

"We had three teams managing four runtimes with six different deployment tools. Nobody had a unified view of what was running"

where."

The Solution

Aether is a Universal Runtime Control Plane that abstracts away runtime differences behind a single, declarative YAML specification.



One Spec, Any Runtime

Write your workload definition once. Aether's Decision Engine automatically selects the optimal runtime based on resource requirements, isolation needs, and infrastructure constraints.



Intelligent Auto-Selection

The built-in Decision Engine evaluates workload characteristics against runtime capabilities to choose the best target. Override it when you need to, trust it when you don't.



Zero-Downtime Migration

Migrate workloads between any two runtimes with a single command. Blue-green and rolling strategies ensure your services stay up during the transition, with automatic rollback on failure.



Built-In Observability

Unified metrics, logging, and health checks across all runtimes. A single TUI dashboard gives operators real-time visibility into every workload, regardless of where it runs.

YAML Spec



Decision Engine



Runtime Adapter



Deployed Workload

Key Metrics

Aether is battle-tested with a comprehensive test suite and a broad feature set designed for production workloads.

4

Runtimes Supported

916

Tests Passing

40+

CLI Commands

37K

Lines of Rust

16

Migration Paths

Zero

Downtime Migrations

Aether covers every permutation of runtime-to-runtime migration with 16 distinct paths, each tested and validated for production reliability.

Runtime Comparison

Aether treats each runtime as a first-class citizen, leveraging the unique strengths of every platform.

Dimension	Podman	Kubernetes	KubeVirt	Metal3
Primary Use Case	Containers & pods on a single host	Orchestrated container clusters	Virtual machines on K8s	Bare-metal provisioning
Scaling Model	Vertical (single node)	Horizontal (multi-node)	Horizontal (VM pools)	Horizontal (hardware fleet)
Isolation Level	Container (cgroups/namespaces)	Container + network policies	Full VM isolation	Physical hardware isolation
Best For	Dev/test, CI/CD, edge	Microservices, stateless apps	Legacy apps, compliance	HPC, GPU, low-latency
Startup Time	Seconds	Seconds to minutes	Minutes	Minutes to hours
Aether Adapter	✓ Full	✓ Full	✓ Full	✓ Full

With Aether, teams no longer need to choose a single runtime upfront. The Decision Engine maps workload requirements to the optimal platform automatically, and migrations between any two runtimes are a single command away.

Migration Strategies

Aether supports three migration strategies, each designed for different risk tolerances and availability requirements.

1. Immediate Migration

1

Stop Source Workload

The existing workload is gracefully stopped on the source runtime with configurable drain timeouts.

2

Deploy to Target Runtime

The workload spec is translated and deployed to the target runtime. Health checks confirm readiness before traffic is routed.

3

Validate & Cleanup

Post-migration validation runs automatically. Source resources are cleaned up after a configurable retention window.

2. Blue-Green Migration

1

Deploy Green Environment

A full copy of the workload is deployed to the target runtime alongside the existing (blue) instance. Both run simultaneously.

2

Switch Traffic

Once the green instance passes health checks, traffic is atomically switched. Rollback is instant if issues are detected.

3. Rolling Migration

1

Incremental Transition

Workload instances are migrated one at a time. Each instance is validated before the next begins, minimizing blast radius.

2

Automatic Rollback

If any instance fails health checks during migration, the entire process is rolled back to the last known-good state.

ROI & Cost Savings

Aether delivers measurable value across three dimensions: infrastructure, operations, and development velocity.

Infrastructure Consolidation

60%

Fewer Deployment Tools

1

Unified Control Plane

4→1

Toolchains Consolidated

Operational Efficiency

75%

Less Context Switching

50%

Faster Incident Response

1

Dashboard for All
Runtimes

Development Velocity

10x

Faster Runtime Migration

80%

Less Boilerplate YAML

Zero

Runtime-Specific Training

Teams adopting Aether eliminate the need for runtime-specific expertise. One abstraction, one set of commands, one operational model for every workload.

Get Started

Three commands to experience the full power of Aether.

Deploy a workload

```
# Deploy to the optimal runtime with a single command
$ aether run --spec workload.yaml

# Aether selects Podman, Kubernetes, KubeVirt, or Metal3
# based on your workload requirements automatically
```

Migrate between runtimes

[illegible]

Observe everything from the TUI

```
# Launch the interactive terminal dashboard
$ aether tui

# Real-time view of all workloads across all runtimes
# Unified logs, metrics, and health checks in one place
```

Ready to Unify Your Runtimes?

Aether is open source, written in Rust, and built for production. Start deploying across Podman, Kubernetes, KubeVirt, and Metal3 today.

Aether — Universal Runtime Control Plane • Built with Rust